

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: a battery cover

Assistant: To better understand your requirements for the battery cover:

1. Could you specify what type of device or system the battery cover will be used in? For example, is it for an electric vehicle (EV), a consumer electronic device (like earphones or fitness equipment), or another application?
2. What is the end use scenario for this battery cover? For instance, is it a structural cover for a large battery pack, a removable cover for a small device, or something else?

Your answers will help me identify the most suitable material options for your application.

User: cover for 100mAh Li-ion battery, it is in a PC

Assistant: Thank you for clarifying the application. To proceed, could you please provide the following details:

1. What is the minimum wall thickness of the battery cover?
2. Do you have any preferred base materials for this cover (such as ABS, PC, or any other polymer)?
3. Are you open to alternative material suggestions if your preferred material is not available or optimal?

These details will help me identify the most suitable material options for your battery cover in the PC.

User: The thickness is 1.5mm I don't have a preferred base material

Assistant: Here are the key material requirements (CTQs) assumed for your battery cover application:

1. Minimum wall thickness: 1.5 mm
2. Processing requirement: Injection molding
3. Flame retardancy: UL94 V-0 at 1.5 mm (halogen-free)
4. Low warpage (dimensional stability)
5. High flow (for thin-wall molding)
6. Adequate stiffness/high modulus
7. Electrical insulation
8. Chemical resistance

Are all of these requirements correct for your battery cover in the PC? Please confirm if these match your needs, or let me know if any should be changed.

User: i also need both hdt and vicat > 150

Assistant: Thank you for confirming the requirements and specifying that both HDT and Vicat should be greater than 150°C.

To finalize the material requirements, could you please clarify the following:

1. Do you have a target Relative Thermal Index (RTI Electrical) for the cover (for example, $\geq 105^{\circ}\text{C}$ or $\geq 130^{\circ}\text{C}$)?
2. Is a high Comparative Tracking Index (CTI) important for your application,

considering the proximity to the PCB?

3. Is Glow Wire Ignition Temperature (GWIT) or Glow Wire Flammability Index (GWFI) required for compliance, or is it not needed for this internal PC cover?

Your answers will help ensure the material suggestions are fully aligned with your needs.

User: i dont have this information recommend grades based on what we discussed

Assistant: Based on your requirements, here are three potential options for your battery cover:

1. CELANEX® XFR 6842 GF30

- UL94 V-0 at 1.5 mm (halogen-free)
- HDT: 210°C, Vicat: 207°C
- High flow for thin-wall molding, high stiffness, excellent electrical insulation
- Good dimensional stability and chemical resistance

2. CELANEX® XFR 6852 GF30

- UL94 V-0 at 1.5 mm (halogen-free)
- HDT: 210°C, Vicat: 207°C
- High stiffness, strong electrical properties (CTI 600 V), good flow

3. CELANEX® XFR6842GF30 ECO-B252

- Same performance as XFR 6842 GF30, plus 25% biobased content (ISCC PLUS mass balance)
- Sustainability benefit with robust electrical and mechanical properties

These grades may be suitable for your application, but final material selection and product design decisions should be based on your specific requirements and compliance needs. Please ensure all applicable standards and regulations are met. Celanese does not manufacture or certify products for critical safety or regulatory applications.

Note: This recommendation is generated by an AI assistant and should be considered purely informational.